

Week 1

This is an outline of the topics we covered in the first week of class.

Monday 8/31

Objectives

- Introduction to the class
- Introduction to graphing solutions of equations.

Introduction to the Course. Welcome to College Algebra!

What are the goals of this course?

Introduction to the People. Who am I?

Who are all of you? Take about 3 minutes to introduce yourself to the people around you. These are going to be your companions on this semester-long journey that we have together. When we come back together, I'll want everyone to introduce themselves with their name and intended major/career path.

Syllabus and Canvas Overview. First off, the syllabus. The entire syllabus is available on our Canvas site.

- Basic info
- Qualifying
- eBook (more on that within Canvas later)
- Attendance
- Student Hours
- Homework
- Quizzes and Tests
- Grade Policy and Final Exam Requirements
- AI and Calculators

For the Canvas site:

- Home page
- Modules
- Agenda page with notes
- Pearson homework and eBook

Graphing. We're going to start this course by talking about graphing. There are many different types of graph, but the one we want to talk about in this class is the kind that allows us to compare two quantities. We want to see how the second quantity changes when the first is modified. For example:

In order to visualize these things, we will need a two-dimensional image, where one direction corresponds to each quantity.

Example 1 Plot and label the points $(2, 1)$, $(-4, 3)$, $(0, 0)$, $(-3, -2)$, $(3, -4)$, and $(2, -1)$.

Solutions of Equations. Now, what do we want to graph? Well, we're going to want to get graphs for mathematical equations (and later, functions). First, we need to look at solutions to these equations. If I have an equation in two variables x and y , then a *solution* is

Example 2 Determine whether each ordered pair is a solution of the equation $3x - 2y = 5$.

(a) $(1, 4)$

(b) $(3, 2)$

Graphs of Equations. What is a graph of an equation?

Graphs of equations that look like $Ax + By = C$ are straight lines. Since it only takes two points to determine a straight line, we can draw these graphs by finding any two points. Two convenient points we can often use are the *intercepts*.

Definition 3 The intercepts of a graph are the places where the graph crosses the axes.

Example 4 Graph $3x - 2y = 5$.

Another way to graph equations of lines is to solve for y first, then plug points into x . In the normal way we will set up these equations in the future, we'll be looking to analyze how y changes when we plug in different values of x , so solving for y is usually the way to go.

Example 5 Graph $5x + 3y = 18$.

When we solve out for y like this, we are indicating that the value of y depends on x , so that y is the *dependent variable* and x is the *independent variable*.

Example 6 Graph $y = x^2 - 3x + 1$.

Proceed to Worksheet 01 on Graphing Equations.